Cover Page

College Name – *Sutex Bank College of Computer Application & Science*

Project Title – *Food Delivery Time Prediction using Machine Learning*

Subject Name – *303 Machine Learning*

Course / Semester – *SYBCA-AI, Semester III*

Student Name – *Shyam Sitapara, Dhruv Vora*

Roll Number – *550, 551*

Faculty – *Hiral A. Patel*

Academic Year – *2026-2027*

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SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS & SCIENCE

All Colleges are Affiliated to Veer Narmad South Gujarat University, Surat.

Self Financed B.B.A.-B.Com. (EM & GM)-B.C.A. (Regular/AI), B.Sc. (DS), M.Sc. (IT), M.Com. (EM & GM) Degree Programme

Accredited by National Assessment and Accreditation Council with "B" Grade

Dr. Mukesh R. Goyani
M.Com., M.Ed., M.Phil., GSET, Ph.D.
Principal

At & Po. : Amroli, Surat,
Pin. 394107, Gujarat (India)

www.amrolicollege.ac.in

(0261) 2494073, 2495643, 2496478
9427113947, 6354626492
principal@amrolicollege.ac.in
Principal_230@vnsgu.ac.in

Certificate

Bachelor of Computer Application (AI)

This is to certify that

Mr. / Ms. **SHYAM JAYESHBHAI SITAPARA (550)**

Exam Seat No. **4597** *has satisfactorily completed*

his/her Practical work in Paper No. **303-05 Fundamentals of Machine Learning (PR)** *for the*

Semester **III** *during the academic year* **2026-27**

Supervisor :

Examiner :

Date : 07/10/26

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principal@amrolicollege.ac.in
Principal_230@vnsgu.ac.in

www.amrolicollege.ac.in

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Mr. / Ms. **DHRUV VIJAYBHAI VORA (551)**

Exam Seat No. **4610** *has satisfactorily completed*

his/her Practical work in Paper No. **303-05 Fundamentals of Machine Learning (PR)** *for the*

Semester **III** *during the academic year* **2026-27**

Supervisor :

Examiner :

Date : 07/10/26

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Principal

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PROJECT PROPOSAL

I. Project Title: - Food Delivery Time Prediction

II. Problem Statement:- *Food delivery services have become an essential part of modern life, where timely delivery is important for customer satisfaction. This project aims to develop a Machine Learning model that predicts food delivery time using factors such as delivery distance, weather conditions, traffic level, vehicle type, preparation time, and courier experience to improve delivery efficiency.*

III. Dataset Source:- Kaggle - Food Delivery Time Dataset

IV. Objectives:-

- To study and understand the food delivery dataset.
- To perform data preprocessing and cleaning.
- To identify missing values and duplicate records.
- To perform Exploratory Data Analysis (EDA).
- To build a Linear Regression model for predicting food delivery time.
- To evaluate the model using R^2 Score, MSE, and RMSE.
- To improve the model using Cross Validation.

V. Expected Output:-

The Machine Learning model will predict the estimated food delivery time based on factors such as delivery distance, weather, traffic conditions, preparation time, vehicle type, and courier experience, helping improve delivery planning and customer satisfaction.

DATASET DESCRIPTION

1. **Dataset Overview:-** The Food Delivery Time Dataset contains information about food delivery orders and the factors that may affect their delivery time. The dataset includes details such as delivery distance, weather conditions, traffic level, time of day, vehicle type, food preparation time, and courier experience. The target variable of the project is Delivery_Time_min, which represents the total delivery time in minutes. The dataset will be analyzed and preprocessed to develop a Machine Learning model for predicting food delivery time.

2. **First 5 Rows:-** df.head()

	Distance_km	Weather	Traffic_Level	Time_of_Day	Vehicle_Type	Preparation_Time_min	Courier_Experience_yrs	Delivery_Time_min
0	7.93	Windy	Low	Afternoon	Scooter	12	1.0	43
1	16.42	Clear	Medium	Evening	Bike	20	2.0	84
2	9.52	Foggy	Low	Night	Scooter	28	1.0	59
3	7.44	Rainy	Medium	Afternoon	Scooter	5	1.0	37
4	19.03	Clear	Low	Morning	Bike	16	5.0	68

3. **Column Names:-** df.columns

```
Index(['Distance_km', 'Weather', 'Traffic_Level', 'Time_of_Day',
       'Vehicle_Type', 'Preparation_Time_min', 'Courier_Experience_yrs',
       'Delivery_Time_min'],
      dtype='str')
```

4. **Dataset Information:-** df.info()

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Distance_km             1000 non-null   float64
 1   Weather                 970 non-null    str
 2   Traffic_Level           970 non-null    str
 3   Time_of_Day             970 non-null    str
 4   Vehicle_Type            1000 non-null   str
 5   Preparation_Time_min    1000 non-null   int64
 6   Courier_Experience_yrs  970 non-null    float64
 7   Delivery_Time_min       1000 non-null   int64
dtypes: float64(2), int64(2), str(4)
memory usage: 83.5 KB
None
```

5. **Statistical Description:-** df.describe()

	Distance_km	Preparation_Time_min	Courier_Experience_yrs	Delivery_Time_min
count	1000.000000	1000.000000	970.000000	1000.000000
mean	10.059970	16.982000	4.579381	56.732000
std	5.696656	7.204553	2.914394	22.070915
min	0.590000	5.000000	0.000000	8.000000
25%	5.105000	11.000000	2.000000	41.000000
50%	10.190000	17.000000	5.000000	55.500000
75%	15.017500	23.000000	7.000000	71.000000
max	19.990000	29.000000	9.000000	153.000000

6. **Dataset Shape:-** df.shape

```
(1000, 8)
```

Row:- 1000

Columns:- 8

7. **Missing Value Analysis:-** df.isnull().sum()

```
Distance_km                0
Weather                   30
Traffic_Level             30
Time_of_Day               30
Vehicle_Type               0
Preparation_Time_min       0
Courier_Experience_yrs    30
Delivery_Time_min          0
dtype: int64
```

8. **Duplicate Value Analysis:-** df.duplicated().sum()

```
0
```

9. **Unique Value Analysis:-** df.nunique()

```
Distance_km               785
Weather                     5
Traffic_Level               3
Time_of_Day                 4
Vehicle_Type                3
Preparation_Time_min       25
Courier_Experience_yrs     10
Delivery_Time_min         108
dtype: int64
```

10. **Data Types:-** df.dtypes

```
Distance_km              float64
Weather                      str
Traffic_Level                str
Time_of_Day                  str
Vehicle_Type                 str
Preparation_Time_min       int64
Courier_Experience_yrs   float64
Delivery_Time_min          int64
dtype: object
```

11. **Numerical Columns:-**

```
Quantitative Data:-  Index(['Distance_km', 'Preparation_Time_min', 'Courier_Experience_yrs',
       'Delivery_Time_min'],
      dtype='str')
```

12. Categorical Data:-

```
Qualitative Data:-  Index(['Weather', 'Traffic_Level', 'Time_of_Day', 'Vehicle_Type'], dtype='str')
```

13. Target Variable:- Delivery_Time_minutes

14. Independent Variables:-

- Distance_km
- Weather
- Traffic_Level
- Time_of_Day
- Vehicle_Type
- Preparation_Time_min
- Courier_Experience_yrs

15. Dataset Summary:-

The Food Delivery Time dataset contains information about food delivery orders and various factors that may affect the delivery duration. It includes numerical and categorical attributes such as delivery distance, weather conditions, traffic level, time of day, vehicle type, food preparation time, and courier experience. The target variable, Delivery_Time_min, represents the delivery time in minutes. Order_ID is used only as an identifier and will not be used as a predictive feature. The dataset will be further analyzed and preprocessed before applying the Linear Regression model to predict food delivery time.

Data Preprocessing

Data preprocessing was performed to clean and prepare the Food Delivery Time Dataset for Machine Learning.

1. Missing Value Handling

The original dataset contained 1,000 records and 8 attributes. Missing values were found in four columns: Weather (30), Traffic_Level (30), Time_of_Day (30), and Courier_Experience_yrs (30). Missing values in categorical columns were replaced using the mode, while missing values in Courier_Experience_yrs were replaced using the median.

After imputation, no missing values remained in the dataset.

2. Duplicate Value Handling

The dataset was checked for duplicate records. No duplicate rows were found.

3. Outlier Handling

Outliers in the Delivery_Time_min column were identified and removed using the Interquartile Range (IQR) method. The lower and upper bounds were calculated using Q1 − 1.5 × IQR and Q3 + 1.5 × IQR, respectively.

4. One-Hot Encoding

The categorical columns Weather, Traffic_Level, Time_of_Day, and Vehicle_Type were converted into numerical indicator columns using One-Hot Encoding. This made the categorical features suitable for Linear Regression.

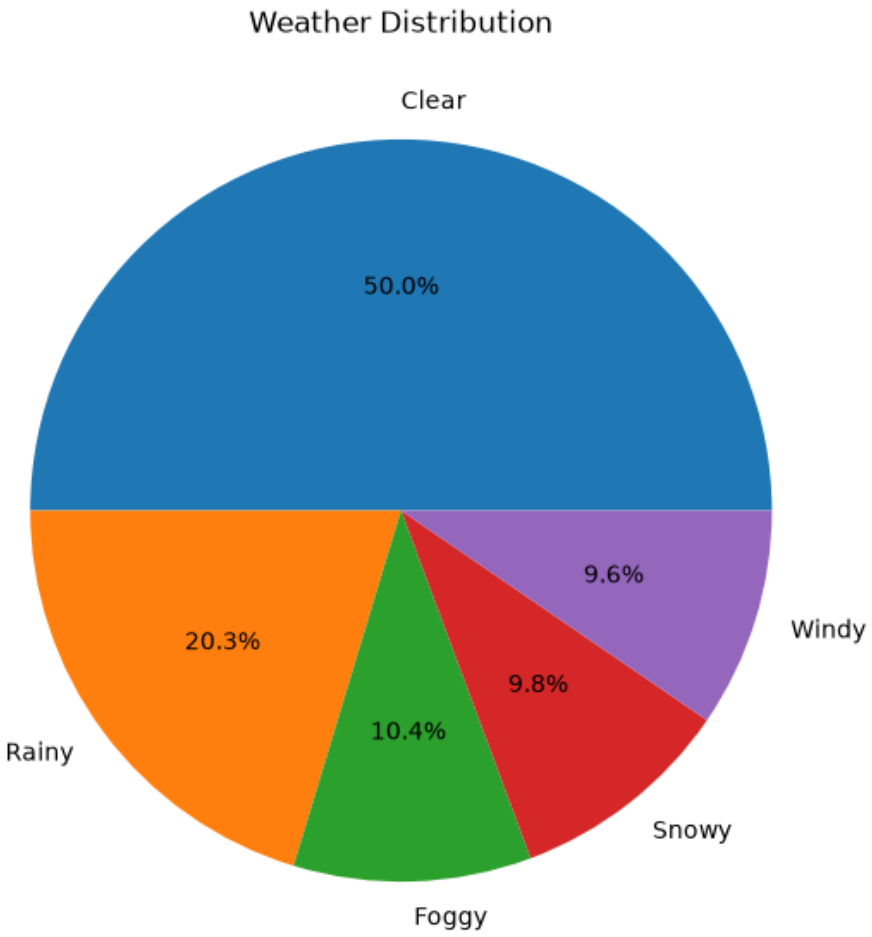
5. Feature Selection

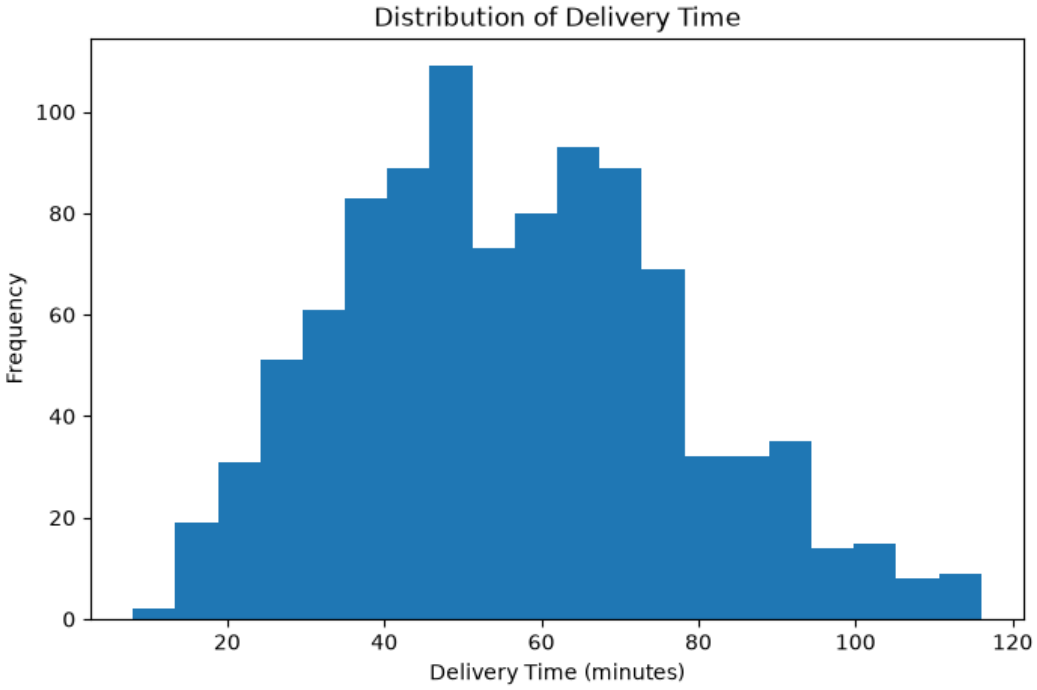
The target variable was Delivery_Time_min. The remaining 18 columns were used as independent variables.

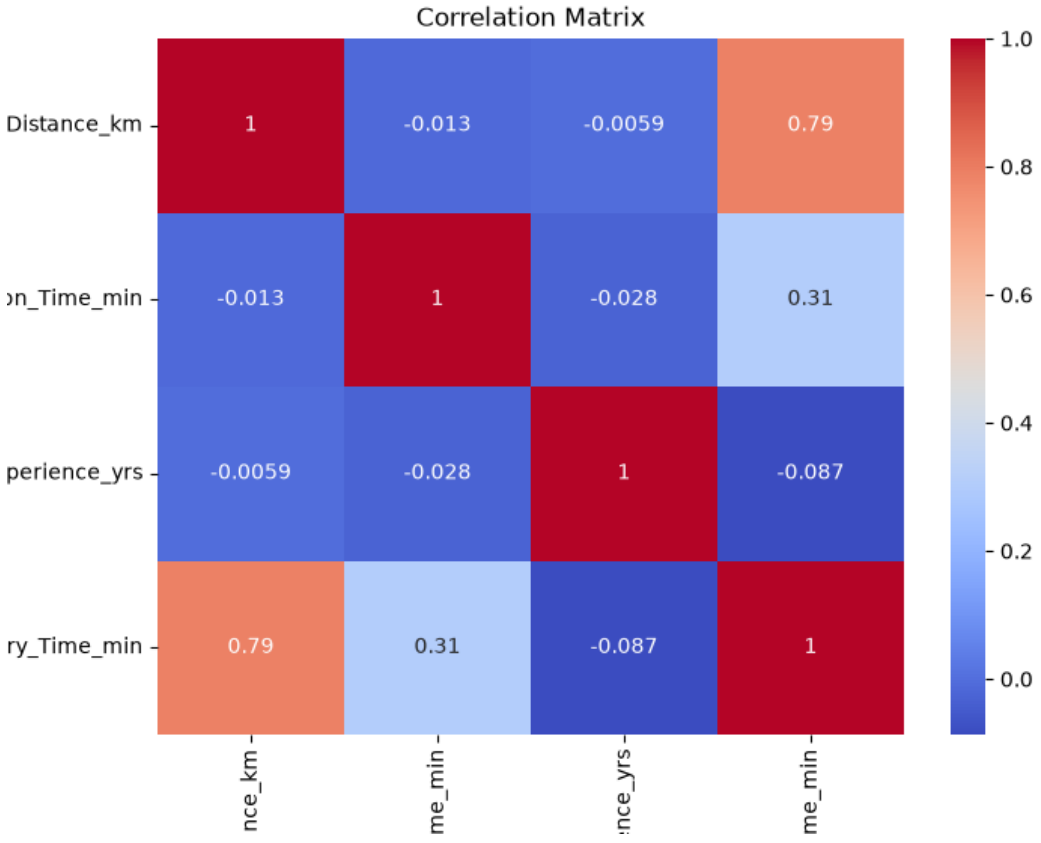
6. Train-Test Split

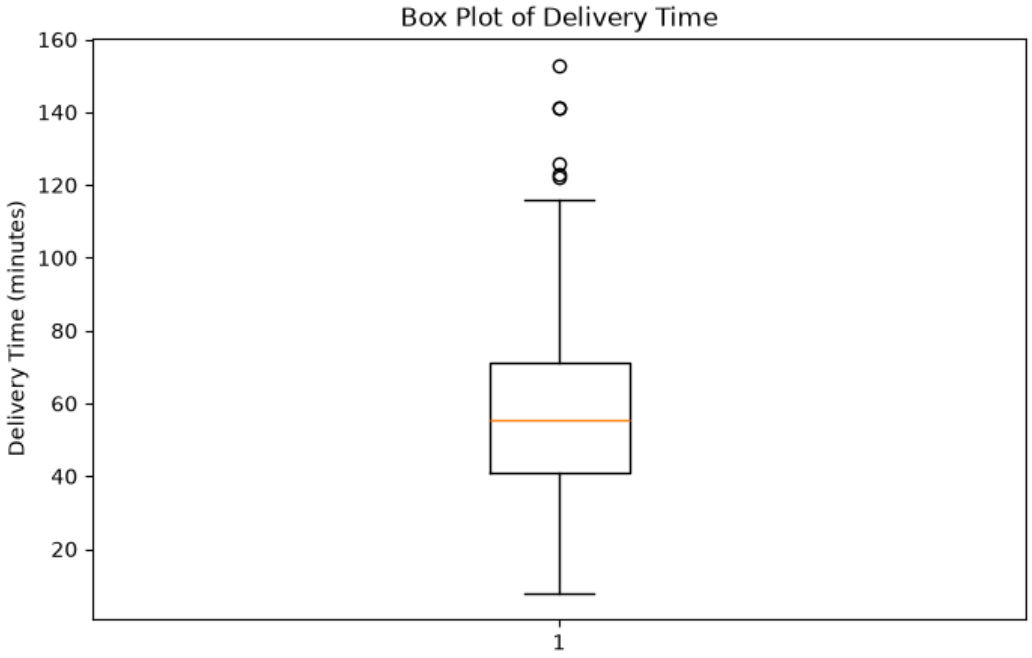
The processed dataset contained 994 records and 18 independent features. The dataset was divided into training and testing sets using an 80:20 split. The training set contained 795 records, and the testing set contained 199 records.

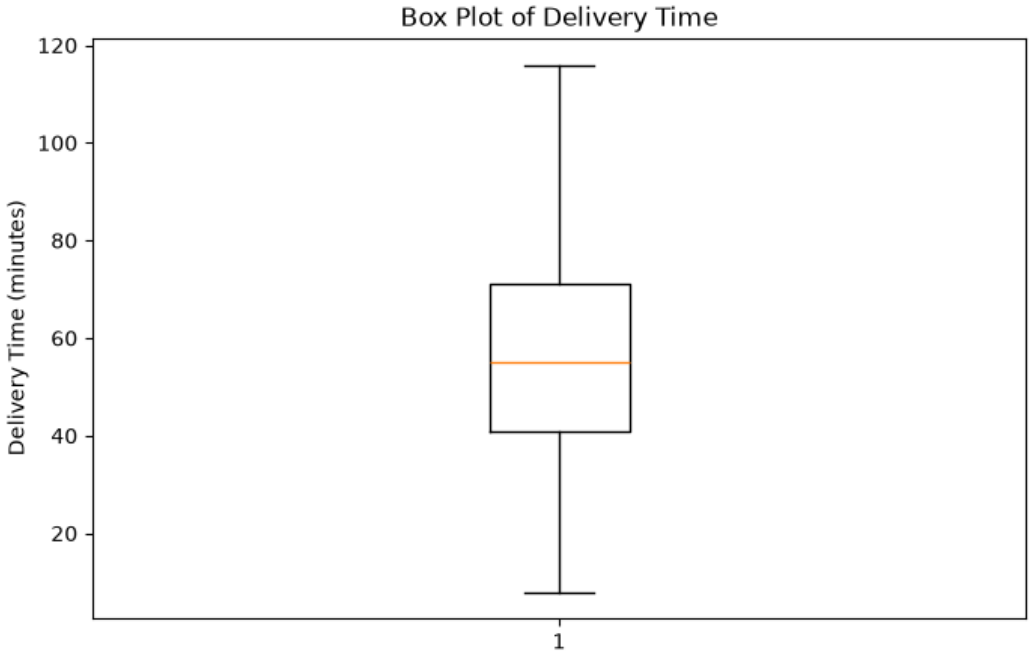
Exploratory Data Analysis

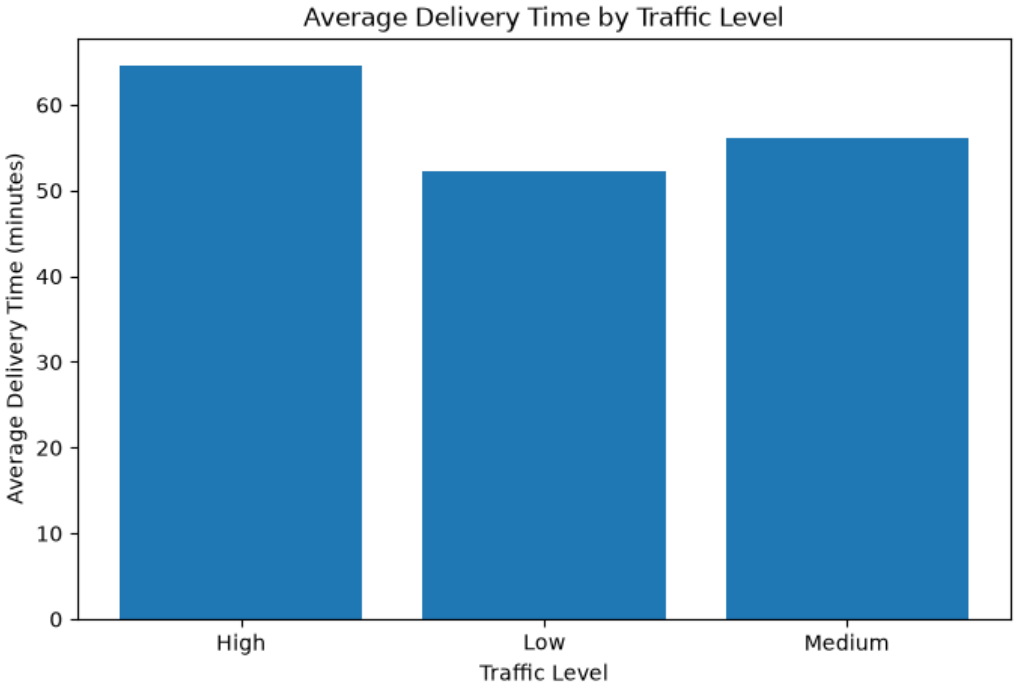



Distribution of Delivery Time
Frequency
0
20
40
60
80
100
20
40
60
80
100
120
Delivery Time (minutes)


Correlation Matrix
Distance_km
ɔn_Time_min
perience_yrs
ry_Time_min
1
-0.013
-0.0059
0.79
-0.013
1
-0.028
0.31
-0.0059
-0.028
1
-0.087
0.79
0.31
-0.087
1
nce_km
me_min
nce_yrs
me_min
1.0
0.8
0.6
0.4
0.2
0.0


Box Plot of Delivery Time
Delivery Time (minutes)
160
140
120
100
80
60
40
20
1


Box Plot of Delivery Time
Delivery Time (minutes)
120
100
80
60
40
20
1


Average Delivery Time by Traffic Level
Average Delivery Time (minutes)
0
10
20
30
40
50
60
High
Low
Medium
Traffic Level

Model Building (Code)

1 Linear Regression

Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values. In this project, it is used to predict food delivery time in minutes based on distance, preparation time, courier experience, weather, traffic level, time of day, and vehicle type.

The dataset was divided into training and testing sets using an 80:20 ratio. The Linear Regression model was trained using the training data and then used to predict delivery times for the testing data.

Python Code:-

```
from sklearn.linear_model import
LinearRegression
```

```
model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

2 Ridge Regression

Ridge Regression is a regularized version of Linear Regression that uses an L2 penalty to help control large model coefficients. It was applied to compare its performance with the Linear Regression model.

Python Code:-

```
from sklearn.linear_model import Ridge

ridge_model = Ridge(alpha=1.0)

ridge_model.fit(X_train, y_train)
```

```
y_pred_ridge = ridge_model.predict(X_test)
```

Both models were evaluated using MAE, MSE, RMSE, and R^2 score.

Model Evaluation

Model evaluation is used to measure the performance of a Machine Learning model. In this project, MAE, MSE, RMSE, and R^2 score are used to evaluate the predictions.

Python Code:-

```
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)
import numpy as np

# Model Evaluation
```

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

Results

The Linear Regression model was evaluated using MAE, MSE, RMSE, and R^2 score. The model achieved an R^2 score of 85.08%, indicating that it explains a good proportion of the variation in food delivery time.

Model Evaluation Results:-

Metric	Result
MAE	5.5792
MSE	73.9204
RMSE	8.5977
R^2 Score	0.8508 (85.08%)

Conclusion

The Food Delivery Time Prediction project was successfully implemented using Machine Learning techniques. The dataset was preprocessed by handling missing values, removing outliers, and converting categorical data into numerical format. Linear Regression was used to predict delivery time based on features such as distance, weather, traffic level, and courier experience. The model achieved an R^2 score of 85.08%, showing good predictive performance on the test dataset. This model can help estimate delivery times and improve delivery planning and customer satisfaction.

Future Scope

In the future, this project can be improved by using advanced Machine Learning algorithms to achieve better prediction accuracy. More real-time data, such as live traffic conditions, weather updates, and restaurant preparation time, can be included to improve predictions. The model can be integrated into a food delivery application to provide customers with accurate estimated delivery times. The system can also be enhanced to handle different cities, peak hours, and special events. In addition, a user-friendly web or mobile application can be developed to make the prediction system more practical and useful.